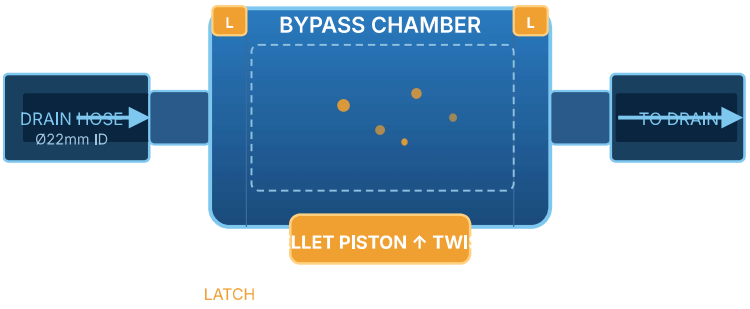
JAMES DYSON AWARD 2026 — TECHNICAL REFERENCE DOCUMENT

Microplastic Filter Snap-On

Design Reference · Parts Breakdown · Dimensions · Assembly



6 parts total
Polypropylene + SS mesh
No tools to install
100 µm pore filtration
Dry pellet ejection
Target cost: <£12

Project: Microplastic Filter Snap-On for Washing Machine Drains | **Document type:** SolidWorks Design Reference | **Revision:** R0.1 — June 2026

Author: Ukacha A. Juma | **University:** Selçuk University — Mechatronics Engineering | **Scale:** NTS (Not to Scale — see dimension tables)

Bill of Materials & Parts Overview

§1 COMPLETE PARTS LIST — 6 UNIQUE PARTS

#	Part Name	Key Dimensions	Material	Qty	Notes
P01	**Main Housing Body**	OD 70mm · ID 64mm · L 140mm Wall: 3mm · Inlet/Outlet port: Ø28mm	PP (Polypropylene)	×1	Bypass chamber. Cylindrical body with two threaded port stubs. Bottom open for piston.
P02	**Snap-On Latch Collar**	OD 34mm · ID 22–25mm stepped L 30mm · Tab width 12mm	PP (Polypropylene)	×2	Inlet + outlet. Dual finger-tab latch. Fits 19–22mm hose OD via stepped inner profile.
P03	**Stainless Steel Mesh Basket**	OD 60mm · L 100mm Pore: 100 µm · Wire: 0.07mm	316 Stainless Steel	×1	Cylindrical basket. Sits inside housing. One end open (water inlet), one end closed (piston face).
P04	**Compression Piston**	OD 61mm · Thickness 18mm Thread: M63×2mm pitch	PP (Polypropylene)	×1	Threaded into housing base. 90° twist compresses fibres. Has central finger grip knob.
P05	**Ejector Cap**	OD 72mm · H 16mm Thread: M70×2mm pitch	PP (Polypropylene)	×1	Threaded cap at base. Removed to eject the compressed pellet. Flat bottom (pellet rests on it).
P06	**EPDM O-Ring Seals**	ID 22mm · Cross-section Ø3mm Standard: BS1806 / AS568A	EPDM Rubber	×2	One per port connection. Prevents leakage at 0.4–0.8 bar drain pump pressure.

Manufacturing note: Parts P01, P02, P04, P05 are injection-moulded PP — ideal for mass production. For prototype: 3D print in PLA (0.2mm layer height, 40% infill for housing, 100% for piston). P03 is sourced: 100 µm woven 316SS mesh sheet, formed into cylinder and spot-welded or crimped.

FIG 1.1 — EXPLODED ASSEMBLY VIEW (SCHEMATIC, NTS)

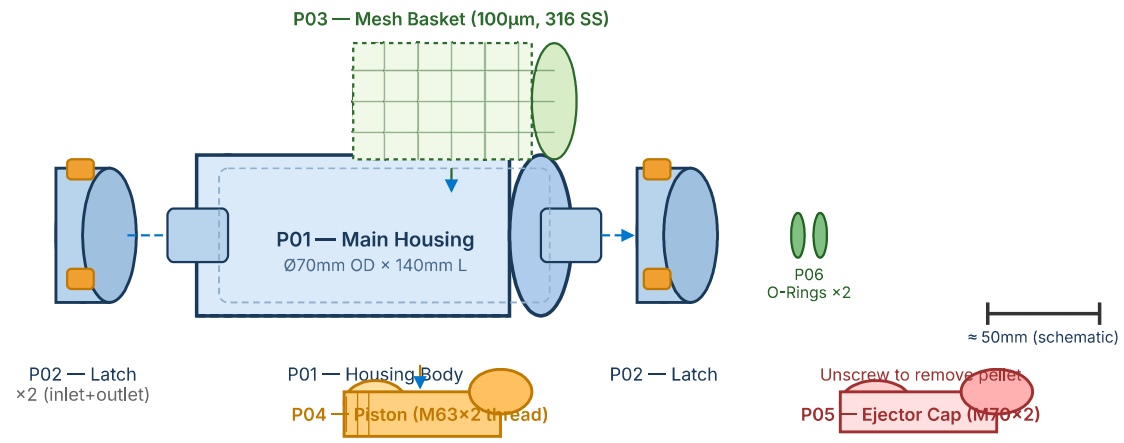


Fig 1.1 — All 6 parts shown in exploded view. Latch collars slide onto ports; mesh basket drops into housing; piston threads into base; ejector cap closes the base.

P01 — Main Housing Body: Dimensions & Drawings

FIG 2.1 — CROSS-SECTION VIEW (FRONT, CUT ALONG AXIS)

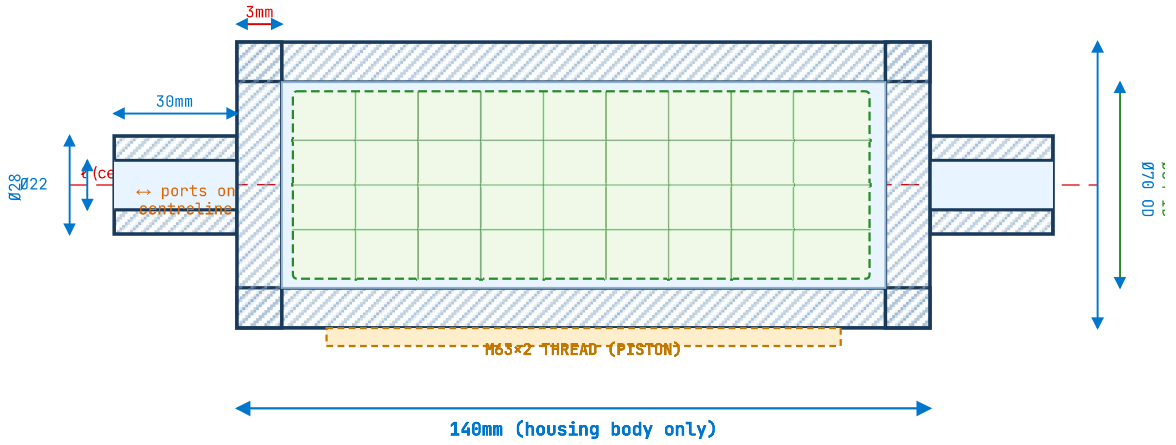


Fig 2.1 — Cross-section showing housing walls (hatched), bypass chamber interior, mesh basket position (dashed green), and port alignment on centreline.

P01 HOUSING DIMENSION TABLE

Feature	Dimension
Outer diameter (OD)	70.0 mm
Inner diameter (ID)	64.0 mm
Wall thickness	3.0 mm
Body length (chamber only)	140.0 mm
Port stub OD	28.0 mm
Port stub ID (bore)	22.0 mm
Port stub length	30.0 mm
Port stub wall	3.0 mm
Total assembly length	~200 mm
O-ring groove depth	1.6 mm
O-ring groove width	3.5 mm
Base thread (for piston)	M63 × 2 pitch
Material	Polypropylene (PP homopolymer)
Surface finish	Ra 1.6 µm (moulded)

▸ FLOW PHYSICS — WHY 70MM?

Velocity reduction calculation:

Typical drain pump flow:

$Q = 20\ L/min = 3.33 \times 10^{-4}\ m^3/s$

At 22mm hose bore:

$v_1 = Q/A = 0.88\ m/s$

At 64mm housing bore:

$v_2 = Q/A = 0.10\ m/s$

Velocity reduction = 8.8× slower
Fibres fall out of suspension below ~0.15 m/s.
The 64mm ID achieves this comfortably. ✅

Pressure drop across housing:

Bernoulli + minor losses ≈ $\Delta P < 0.05\ bar$
Well within pump tolerance (faults above 0.3 bar back-pressure). ✅

⚠ **SolidWorks note:** Model the housing as a single extruded body. The port stubs are *boss extrusions* on the end faces. The internal bore is a *cut extrude* from each end meeting in the middle. Use *Shell* feature (3mm thickness) for the cylindrical walls.

P02 Latch Collar · P03 Mesh Basket · P04 Piston

FIG 3.1 — P02 SNAP-ON LATCH COLLAR (CROSS-SECTION)

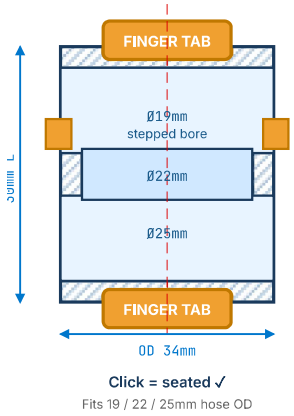


FIG 3.2 — P03 MESH BASKET (SECTION + END VIEW)

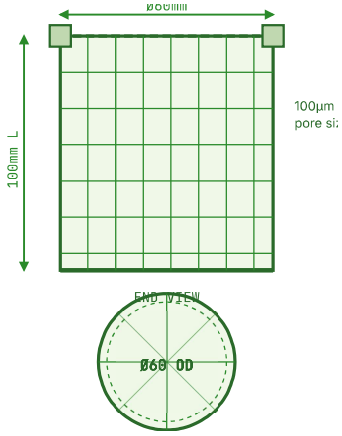
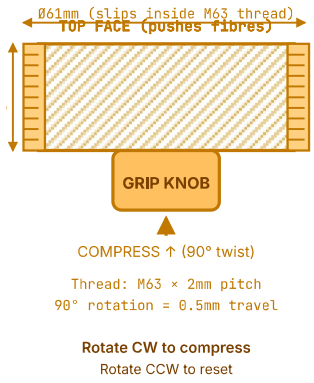


FIG 3.3 — P04 COMPRESSION PISTON (CROSS-SECTION)



Part	OD	ID / Bore	Length / Height	Thread	Material
P02 — Latch Collar (×2)	34 mm	19 / 22 / 25mm stepped	30 mm	—	PP
P03 — Mesh Basket	60 mm	Ø56mm ID	100 mm	—	316 SS, 100µm
P04 — Compression Piston	61 mm	—	18 mm disc + 28mm knob	M63 × 2	PP
P05 — Ejector Cap	72 mm	Ø64mm ID	16 mm	M70 × 2	PP
P06 — O-Ring (×2)	28 mm (OD)	22 mm (ID)	3 mm cross-section	—	EPDM

Assembly Sequence & Operation Diagram

§4 STEP-BY-STEP ASSEMBLY

FIG 4.1 — INSTALLATION & ASSEMBLY SEQUENCE (5 STEPS)

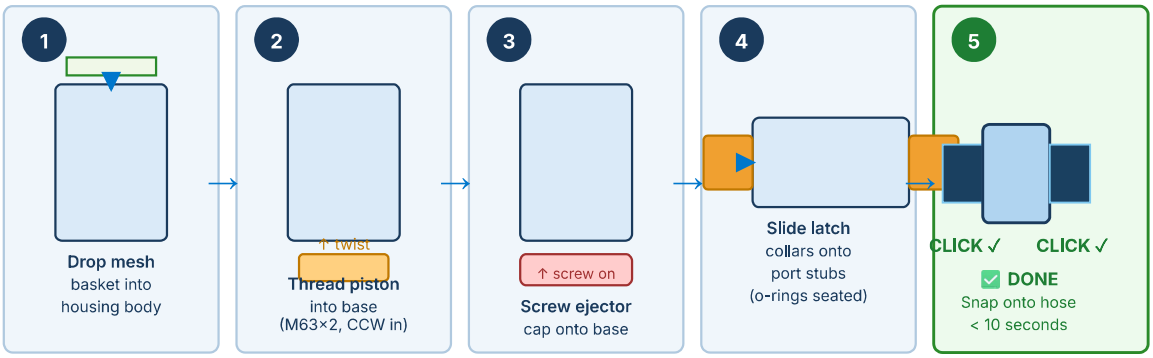


Fig 4.1 — Full assembly in 5 steps. Steps 1–4 done once; Step 5 is the daily-use snap-on to the machine.

§5 OPERATION CYCLE — HOW IT WORKS IN USE

FIG 5.1 — OPERATIONAL FLOW DIAGRAM (DURING WASHING MACHINE DRAIN CYCLE)

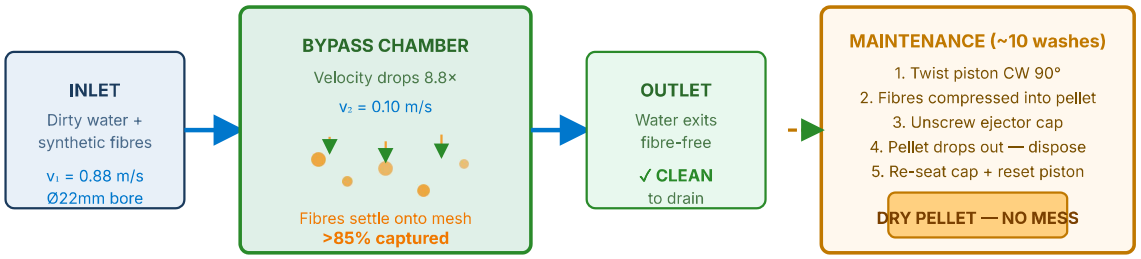


Fig 5.1 — Operational flow. Water pressure from the drain pump drives the cycle. Zero electricity needed. Maintenance is needed approximately every 10 wash cycles.

💡 **SolidWorks modelling order (recommended):**

1. Housing body (P01) — shell feature, 3mm wall
2. Port stubs — boss extrude on end faces
3. O-ring groove — revolve cut on stub OD
4. Piston (P04) — revolve, add thread cosmetic feature
5. Ejector cap (P05) — revolve
6. Mesh basket (P03) — surface body or thin-shell revolve
7. Latch collar (P02) — extrude + cut for bore steps
8. Assembly — mate all parts, add flow simulation

⚠ **Critical tolerances to set:**

- Piston to housing: **H7/f6** fit (sliding clearance ~0.05mm)
- Mesh OD to housing ID: **0.5mm radial clearance** (allows easy drop-in)
- O-ring groove: squeeze 15–20% of cross-section (standard EPDM)
- Latch tab snap: **0.3mm interference** on the catch — audible click
- Thread engagement: minimum **3 full turns** (M63×2 → 6mm engagement)

SolidWorks Modelling Guide & Material Specs

§6 SOLIDWORKS PART MODELLING SEQUENCE

Part	SW Feature Sequence	Key Sketch Dimensions	Notes
P01 Housing	Revolve Boss → Shell (3mm) → Extrude port stubs → Cut bore through stubs → Revolve cut O-ring groove → Revolve cut thread at base	R35 outside, R32 inside, L=140mm body	Start with revolved profile. Shell removes interior. Stubs added as boss extrudes on flat end faces.
P02 Latch	Revolve base cylinder → Extruded cut stepped bore → Extrude tabs (2×) → Fillet tabs 2mm → Cut latch catch groove	R17 OD, stepped bore 9.5 / 11 / 12.5mm, L=30mm	The stepped bore inner profile must be done as a revolve cut. Tab geometry is a simple rectangle extrude on the OD face.
P03 Mesh Basket	Revolve thin shell (0.5mm wall) → Cap one end → Add rim flange	R30 OD, R28 ID, L=100mm	Use Surface → Revolve for the thin mesh wall. In real life this is formed SS mesh — in SW treat as a thin-walled cylinder. Apply Weld Bead cosmetic feature for the seam.
P04 Piston	Revolve disc → Extrude grip knob boss → Cut M63 cosmetic thread	R30.5 disc, H=18mm, knob H=28mm R12	Use Insert → Annotations → Cosmetic Thread for the M63×2 thread. Physical thread modelling is not required for FEA or Flow Sim.
P05 Ejector Cap	Revolve flat cap disc → Extrude rim walls → Cut M70 cosmetic thread	R36 disc, rim H=12mm, wall 3mm	Simple cup shape. Critical fit: cap ID must match housing OD at base within 0.1mm.
P06 O-Ring	Revolve torus: path circle R11, profile circle R1.5	R11 centreline, R1.5 cross-section	Standard BS1806-222 o-ring. Can use downloaded STEP file from Parker or Trelleborg.

§7 MATERIAL PROPERTIES FOR SIMULATION

Property	Polypropylene (PP Homo)	316 Stainless Steel	EPDM Rubber
Density	905 kg/m³	7,990 kg/m³	1,150 kg/m³
Young's Modulus	1.5–2.0 GPa	193 GPa	~0.003 GPa
Tensile Strength	30–40 MPa	515 MPa	10–15 MPa
Chemical resistance	Excellent (detergents, hot water)	Excellent (all conditions)	Excellent (hot water)
Max continuous temp	100°C (PP-H)	870°C	150°C
SW Material Library	Plastics → PP	Steel → 316 SS	Rubber → EPDM

§8 SOLIDWORKS FLOW SIMULATION SETUP

Boundary Conditions:

- Inlet: Volume flow rate = `3.33×10⁻⁴ m³/s` (20 L/min)
- Outlet: Static pressure = `0 Pa` (atmospheric)
- Walls: No-slip condition

Static Stress Analysis (FEA):

- Fix housing body at port face
- Apply 0.5 bar internal pressure to chamber walls
- Apply 20N point load on each latch tab (finger press force)
- Check: factor of safety >3 on PP (yield = 30 MPa)

- Fluid: Water at 60°C (hot wash)
- Turbulence: k-ε model

Goals to track:
- Velocity at mid-chamber cross-section (target: <0.15 m/s)
- Pressure drop inlet→outlet (target: <0.05 bar)
- Surface pressure on latch tabs (target: <0.3 bar)

Expected results:
- Max stress on latch tab ~8 MPa → FoS = 3.75 ✅
- Housing wall stress ~2 MPa → FoS = 15 ✅
- No yielding anywhere at operating conditions ✅

§9 PROTOTYPE SOURCING (FOR 3D PRINT)

Item	Specification	Source / Notes	Est. Cost
PLA filament	`1.75mm, white or grey`	Any 3D print supplier	~£5
316SS mesh sheet	`100 µm pore / 0.07mm wire, A4 sheet`	eBay, Amazon, RS Components	~£8
EPDM o-rings (×2)	`22mm ID × 3mm CS (BS1806-222)`	RS Components, Farnell	~£2
Total prototype cost			**~£15**

💡 **Print settings for housing (P01):** 0.2mm layer height, 40% gyroid infill, 4 perimeter walls (for pressure resistance), PETG preferred over PLA for hot water resistance. Print port stubs oriented vertically for better bore surface quality.